

A transfer reinforcement learning and digital-twin based task allocation method for human-robot collaboration assembly

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ABSTRACT

Currently, human-robot collaboration systems are considered to have great application potential in complex and flexible assembly tasks. In human-robot collaborative assembly systems, the roles and tasks allocation between human and robot are the vital stage to exert and integrate the strength of both. Current many researches of task allocation mainly train decision-making model depending on a large number of predefined standard assembly data such as assembly time and assembly difficulty. However, due to individual differences and frequent product updates, the dynamic assembly conditions lack enough data for model training, and the constructed model based on historical data may not adapt to the new condition. To solve this gap, a transfer reinforcement learning and digital-twin based task allocation method is proposed to achieve the accurate and efficient multi-agent human-robot collaboration task allocation policy learning. Firstly, based on the digital twin of human-robot collaboration environment, the augmented reality is leveraged to simulate assembly process and collect execution data of workers and robots before the physical assembly. Secondly, a multi-agent reinforcement learning method is introduced to use domain randomization strategy to pre-train decision policy before the accurate model training. Thirdly, the knowledge distillation is leveraged to develop the transfer reinforcement learning framework to learn the accurate decision policy by reusing the pre-trained policy and utilizing the collected assembly simulation data. Finally, a case study is performed to demonstrate the effectiveness of the proposed method.

1. Introduction

In the intelligent manufacturing system, the human-robot collaboration (HRC) has been considered as the potential solution for complex production tasks, which have been received much attention in recent years. Some studies reported that human-robot collaboration mode has excellent performance in terms of efficiency and flexibility by integrating the cognition of humans and precision of the robots (Wang et al., 2019; Müller et al., 2016). In a HRC system, the basic mode of multiple entities collaboration is that each entity performs different subtasks of a common task or product (Li et al., 2023a). Due to the differences between humans and robots, allocating and scheduling subtasks between the two is the vital stage of HRC, which determines whether they can exert and integrate their respective strength completely (Lu et al., 2022). Therefore, the operating capabilities and current status of the human and robots must be considered for task allocation (Malik and Bilberg, 2019a). In addition, the human operation difficulty and work fatigue should be considered in the task allocation, to avoid the quality decrease by the high workload (Merlo et al., 2023).

Many studies established mathematical models for HRC task planning and allocation problems and used optimization algorithms to solve the optimal allocation scheme. Zhu and Huang et al. (Zhu et al., 2022) proposed a digital twin-driven HRC manufacturing reconfiguration optimization model, and utilized the NSGA-II algorithm to optimize the configuration of HRC system. Ren et al. (2023) constructed the joint optimization model that consider the complete time and total production costs minimization to solve the assembly operation sequence and the task allocation for HRC system. However, limited by the calculating speed, the optimization algorithm cannot be applied to dynamic and complex decision-making scenarios.

With the development of the artificial intelligence methods, many researches of recent years focus on utilizing the deep learning (DL) and reinforcement learning (RL) methods to acquire the task allocation and schedule policy (Yu et al., 2020, 2021; Zhang et al., 2022; Song et al., 2022; Zheng et al., 2022). Using DL-based methods can acquire decision-making strategies, knowledge or experience before the deploying, and provide decision-making results efficiently at the manufacturing site. For example, Li et al. (2023b) used the temporal

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subgraph reasoning method to plan the robots and human tasks with the constructed knowledge graph of the assembly process. Reinforcement learning, especially multi-agent reinforcement learning, has good sequential decision-making learning capabilities (van Otterlo and Wiering, 2012), and is very suitable for addressing HRC task allocation problems. Zhang et al. (2022) proposed a multi-agent reinforcement learning based method for human-robot assembly task sequencing and allocation. Yu et al. (2021) developed a deep-Q-learning (DQN)-based multi-agent reinforcement learning method to optimize the task scheduling of HRC. Liu et al. (2021) proposed an adapted training approach based on the DQN, which takes into account the uncertainty of human for HRC task allocation. These learning methods and optimization methods are all used to obtain decisions relying on execution data such as assembly time, operation difficulty and human workload. However, current manufacturing systems lack sufficient accurate execution data for newly changed work conditions, particularly for new products and new workers. This is due to the limited time for data accumulation and the cost constraints associated with real-world assembly evaluation experiments, which hinders effective decision model training. Moreover, due to the condition changes from products and workers, the trained decision-making models in historic data may not be suitable in practice.

In order to address this gap, this study proposed a transfer reinforcement learning and digital-twin based task allocation method for human-robot collaboration assembly. Before the physical assembling on site, the AR and digital-twin based assembly environment are leveraged to efficiently collect the individual assembly execution data. The multi-agent reinforcement learning algorithm is developed to pre-train the decision policy using the domain randomization before the targeted decision policy learning. And then constructing the transfer reinforcement learning framework based on the knowledge distillation, the multi-agents can learn the accurate decision policy for new working conditions at a low time cost by utilizing the collected execution data and reusing the pre-trained decision experience.

This paper is organized as follows. In section 2, related work is presented while in section 3, an AR and digital twin based HRC assembly environment is introduced, defining the required operational data and acquiring method for these data. In section 4, a transfer reinforcement learning framework for HRC task allocation is described, followed by the relevant functions are defined and pre-train process are introduced. Then, in section 5, shows a demonstration of the proposed transfer reinforcement learning method in a AGV steer wheel assembly, which evaluates the performance of training and decision-making for multiple human and robotic agents. Finally, section 6 concludes this study and highlights future research directions.

2. Related work

2.1. Task allocation of human-robot collaboration in manufacturing

With the current popularity of collaborative robots, the human-robot collaboration model has attracted a lot of attention and is regarded as a production model with potential. Many studies on HRC strategies and methods have been proposed, among which HRC task allocation has been the focus of research (Matheson et al., 2019). Allocating subtasks to humans and robots reasonably is very necessary, and an optimized subtasks allocation scheme can give full play to their respective capabilities and achieve more efficient assembly.

For the traditional methods, the execution complexity and the potential for automation of subtasks are intuitively analyzed, and allocated based on human and robot capabilities. The executing complexity of the assembly tasks is assessed through part structure (Malik and Bilberg, 2019b), task description (Malik and Bilberg, 2019a), and functional requirements (Mateus et al., 2019) and then subtasks are assigned to humans or robots based on the evaluated complexity. In addition to the capability constraints, maximizing objectives, such as time, cost, and quality, are also the vital objectives in the aspect of HRC task allocation

decision (Xu et al., 2020; Liu et al., 2018; Lee et al., 2022). Many researches utilized assumptions and mathematical models to abstractly model the HRC task sequence optimization and allocation problems, and then solved these problems by developing optimization algorithms such as particle swarm optimization algorithm (PSO), genetic algorithm (GA) (Liau and Ryu, 2020). Banziger et al. (Bänziger et al., 2020) developed simulation tools to calculate different target parameters such as production time, and then utilized a genetic algorithm to optimize task allocation for HRC with simulated target parameters. Lee et al. (2022) presented a disassembly sequence planning (DSP) models considering resources and safety constraints in the HRC with aiming to minimize the total disassembly time. Furthermore, in order to reduce human workload, ergonomic principles also need to be taken account into human-robot collaborative distribution methods. Petzoldt (Petzoldt et al., 2022), Gualtieri (Gualtieri et al., 2023) and E. Lamon et al. (2019) considers the human fatigue, human state and skill levels in the HRC task allocation framework, to make the task allocation consistent with ergonomic principles. Gang Yuan et al. (2023) proposed a novel heuristic algorithm to comprehensively evaluate the resilience of human-robot collaborative disassembly system from the aspects of efficiency, technical, economic and environments.

In summary, these methods based on optimization algorithms are limited by solution efficiency and fixed models, and thus focused on single or fixed tasks. However, with online planning and real-time decision-making of robots, the flexible HRC should be suitable for dynamic and multiple HRC optimization problems, and the assignment and scheduling of HRC tasks should be performed in an intelligent manner (de Gea Fernández et al., 2017).

2.2. Learning based methods for human robot collaboration decision-making

Compared with traditional manufacturing decisions, relevant decision-making in HRC scenarios need to consider human intentions and state, predict uncertainty caused by human, and provide decision results quickly and efficiently to guide the operation of the HRC system in real time. Traditional optimization methods are difficult to describe these factors, but learning-based methods can learn and understand more complex HRC scenarios and support intelligent decision-making. Therefore, many recent studies have focused on more intelligent and flexible methods to solve the task planning and allocation problem of HRC scenarios by applying intelligent methods including deep learning and reinforcement learning.

Deep learning has been used in many studies to understand the scenarios and then assist in decision-making (Wang et al., 2019). Scene graph (SG) is a widely used HRC scene description method, and SG can be used as HRC state and contextual information to support task assigning and planning (Fan et al., 2022). In order to acquire the SG, Hassam Riaz et al. (2020) presented a multi-level scene description neural network to capture scene graph of HRC. Li et al. (2022) utilized scene graph to learn scenario interpretation for on-site collaborative work. Moon et al. (Moon and Lee, 2018) utilized GCN-based 3D semantic graph mapping to generate scene descriptions. In addition, knowledge graphs can also be used for describing information in human-robot collaboration scenarios (Zheng et al., 2022). Then, HRC task planning and allocation are implemented and assigned to human and robot agents through real-time perception and temporal subgraph reasoning. Reinforcement learning algorithms can be directly used to train decision-making strategies for solving complex HRC task allocation problems without domain knowledge guidance. Nikolakis et al. (2018) utilized a hierarchical model to represent resources and workloads, and then utilize intelligent search to adjust and schedule new task allocations online. Yu et al. (2020) utilized RL algorithms based on MCTS and CNN to generate work sequences in HRC systems by considering some specific rules. Furthermore, in order to consider the differences in the capabilities of humans and robots, Zhang et al. (2022) established a

method based on multi-agent reinforcement learning to plan sequence and assign of HRC assembly tasks.

In general, the learning methods require sufficient and accurate data to train the model. Although reinforcement learning obtains training data through the interaction between the agent and the environment, the establishment of the training environment also needs to be supported by reasonable data from real scene. However, the current manufacturing system lacks sufficient data for new working conditions such as new product manufacturing processes, new models of robots, or new employees. Moreover, in order to reduce the time taken to put the method into practice, the efficient training method is expected to enhance the adaptability of agents to new tasks (Raziei and Moghaddam, 2021).

2.3. Digital twin driven human robot collaboration

Digital twin technology is one of the core technologies of Industry 4.0, and the proposed digital twin workshop with five dimensions can serve as a new paradigm of Industry 4.0 (Tao et al., 2018). In HRC researches, digital twins are often used as a driving technology. For example, Dallel et al. (2023) use digital twin and virtual reality to obtain human activity data in human-robot collaboration system for training human actions recognition networks. Bilberg et al. (Bilberg and Malik, 2019) discussed an object-oriented event-driven digital twin, which is a flexible simulation unit coordinated with HRC, used to dynamically balance workload between humans and robots. Similarly, the HRC assembly framework based on digital twins also proposed by Lv et al. (2021), and utilized to improve the overall assembly efficiency and ensure the safety. Liu et al. (2022) and Shi et al. (2022) investigated the cognitive digital twin method and framework for HRC scene to promote cognition of human-centric HRC. Furthermore, many DT-based methods build HRC collaboration systems by connecting the physical and virtual space. Malik et al. (Malik and Brem, 2021) used DT to construct collaborative production systems with high complexity, which achieved resource sharing along the system's life cycle by modelling physical-digital space connection of assembly tasks in each phase and corresponding environment, components, and task allocation.

An important task of digital twin technology is to exploit abstract knowledge and useful data, while analyze and process these data (Jiang et al., 2021). For learning-based scheduling methods, utilizing digital twins can provide necessary data to support decision models training. In addition, digital systems and digital spaces based on digital twins can

also provide support for process and system simulation (Schluse et al., 2018; Phanden et al., 2021), allowing product data to be obtained before there is a physical entity. Therefore, in the case of new products or new employees, digital twin simulation can provide useable data to support application decisions. Furthermore, models trained on historical empirical data can be combined with new simulation data to train available decision-making models at a lower cost.

3. Digital twin based human robot collaboration assembly

3.1. Digital twin of human robot collaboration assembly system

A typical human robot collaboration system consists of one work, one robot, one work station, some tools and other components to complete a set of specific assembly tasks. In this system, the human and robot share the same work space and operating tools, and accomplish together a set of given assembly tasks. Therefore, as shown in Fig. 1, the corresponding HRC digital twin system consists of: digital twin human model that can follow the movements of real workers seamlessly through real-time human motion data from motion capture equipment; digital hand manipulation that can map user's finger-level hand movements based on hand gesture data tracked by Hololens2; digital twin robot that could move and operate according to the real-time data from the real robot controller; and other digital components that can be affected by the digital human and robot in the way of physical simulation engine. Due to the states mapping from real space to the digital twin, analysis of states can be performed in the digital space, such as task accessibility and visibility for the worker (see Fig. 2).

For HRC assembly system, the task allocation problem involves reasonably allocating a set subtasks of the product to humans or robots with the multiple optimization objectives: minimize assembly time, work difficulty, and human worker fatigue. A subtask can be performed by a human alone, by a robot alone, or by both in collaboration. Further, for a collaborative subtask, it is decomposed into operation involving parts (such as installation) and operation involving tools (such as tightening), and different operations should be assigned to collaborative humans or robots. At the same time, due to capacity limitations, some tasks or operations cannot be performed by humans or robots. The execution order of these subtasks to be assigned is determined based on the assembly process, which depends on the structure and assembly constraints of the product.

The task allocation model is trained to allocate tasks based on the

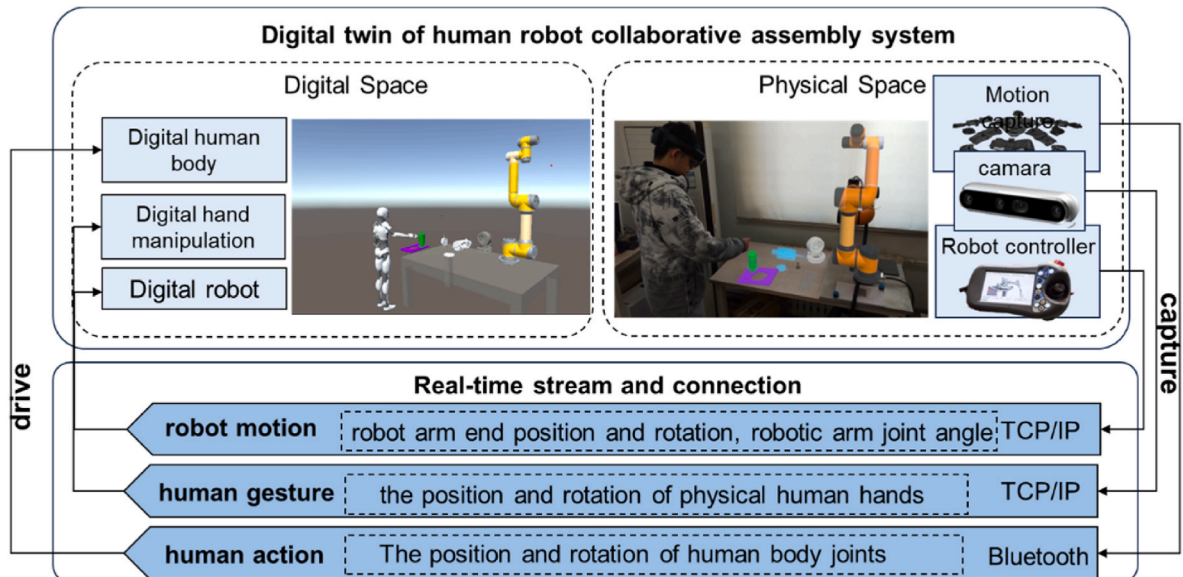
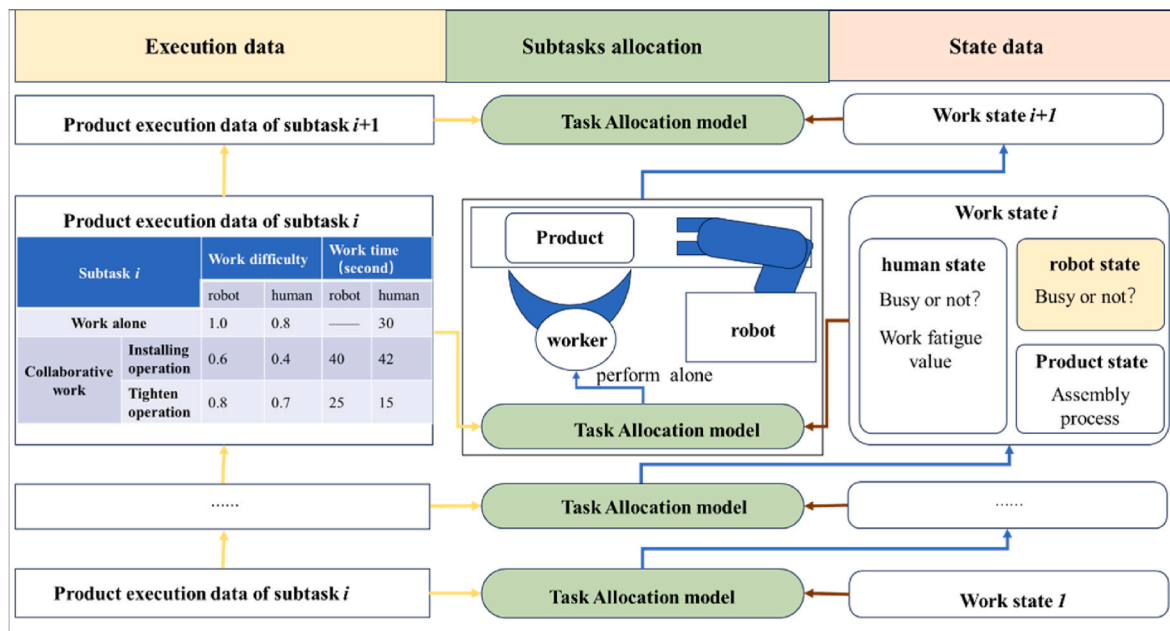


Fig. 1. Digital twin of human robot collaboration assembly system.



necessary information with the optimization objectives. Task allocation decision requires two aspects of data involving the execution data and the state data. The execution data involves the difficulty and time of executing tasks by agents, which represents the capability of the human and robots for performing task. The working state of HRC system is also needed, including the busy state of both, fatigue of human and work process, which represents the available status of human and robot. The assigned tasks are synchronized to the AR device, and then the worker is guided to complete the task through the assistance of the AR device. For robots, the corresponding control instructions of the task are executed.

The execution data includes the execution difficulty and time of the subtask being executed alone, and of the subtask being executed collaboratively. In the situation of collaborative execution of subtask in our study, the collaborative subtask can be split into two parts including installation operation and tightening operation, and each one need to be performed separately by humans or robots. The execution data is shown in Eq. (1).

Where E_i^j denotes the execution data of subtask i , and j represents the related assembly component of this subtask. Except for collaborative execution, only one individual is allowed to assemble on the assembly part at the same time, due to work space and safety constraint. For the subtask can be completed by robot or human independently, the D_A and T_A denotes the execution difficulty and time of performing subtask alone. Moreover, some subtasks can be completed by collaboration, D_I and T_I represent the execution difficulty and time for installing operation of subtask, and D_T and T_T represent the execution data for the tightening operation of same subtask. In the array, D^H and D^R respectively represent the difficulty of humans performing tasks and the difficulty of robots performing tasks, and T^H and T^R respectively represent the needed time of humans performing tasks and of robots performing tasks. If the difficulty and time values are absent, it means that the robot or human cannot perform the task or operation alone. For example, $E_8^1 = \{(0.6, 1, 42, -), (0.5, 0.6, 30, 24, 0.6, 1, 18, -)\}$ represents that the task 8 can't be executed by robot alone, and the operation involving tools also

Execution data will change due to dynamic changes in the working conditions of human-robot collaboration, such as product update iterations, improvements in worker proficiency, replacement of more flexible robots. In the process of acquiring and analyzing the execution data automatically, although the movements of humans and robots can be captured through the digital-twin, capturing the real-time state of physical assembly entities is difficult and in a high cost. In order to solve the problem, an assembly simulation platform based on AR devices and HRC digital twin system is developed, as shown in Fig. 3. As reported in our previous study (Mao et al.), users can simulate the entire assembly process without the physical product in this platform. By using the 3D display and interactive interface provided by AR device, the user can manipulate virtual parts to execute subtasks in real workstation. The virtual robot is driven by the position data from real robot controller program to execute subtasks. In this process, because the human and real controller participates in simulation, the real-time data streaming and connection from DT of HRC system are used to drive the virtual human and robot models of DT space. Compared with the physical assembly, the real-time assembly data, including human actions, assembling parts state and robot arm movement, are more easily collected from the digital space. The collected data are then leveraged to analyze the subtask execution data.

Execution time can be acquired through simulation of subtask operations. For humans, execution difficulty can be broken down into accessibility and ergonomics. In our method, the operation accessibility index of human (ORI) is used to estimate the accessibility of the assembly operation, and this ORI is related to the max distance of user head to the assembling parts in the assembly step operation process. According to the research (Enomoto et al., 2013), the distance including four levels: less than D1, between D1 and D2, between D2 and D3, greater than D3, and these values are determined by the height of the operator. Corresponding to these numerical levels, the four reachability are as follow: easy, requiring effort, difficulty and impossible. The Ergonomic index (EI) is proposed to evaluate the actions of assembly operation from the ergonomics, and this index is the weighted sum of

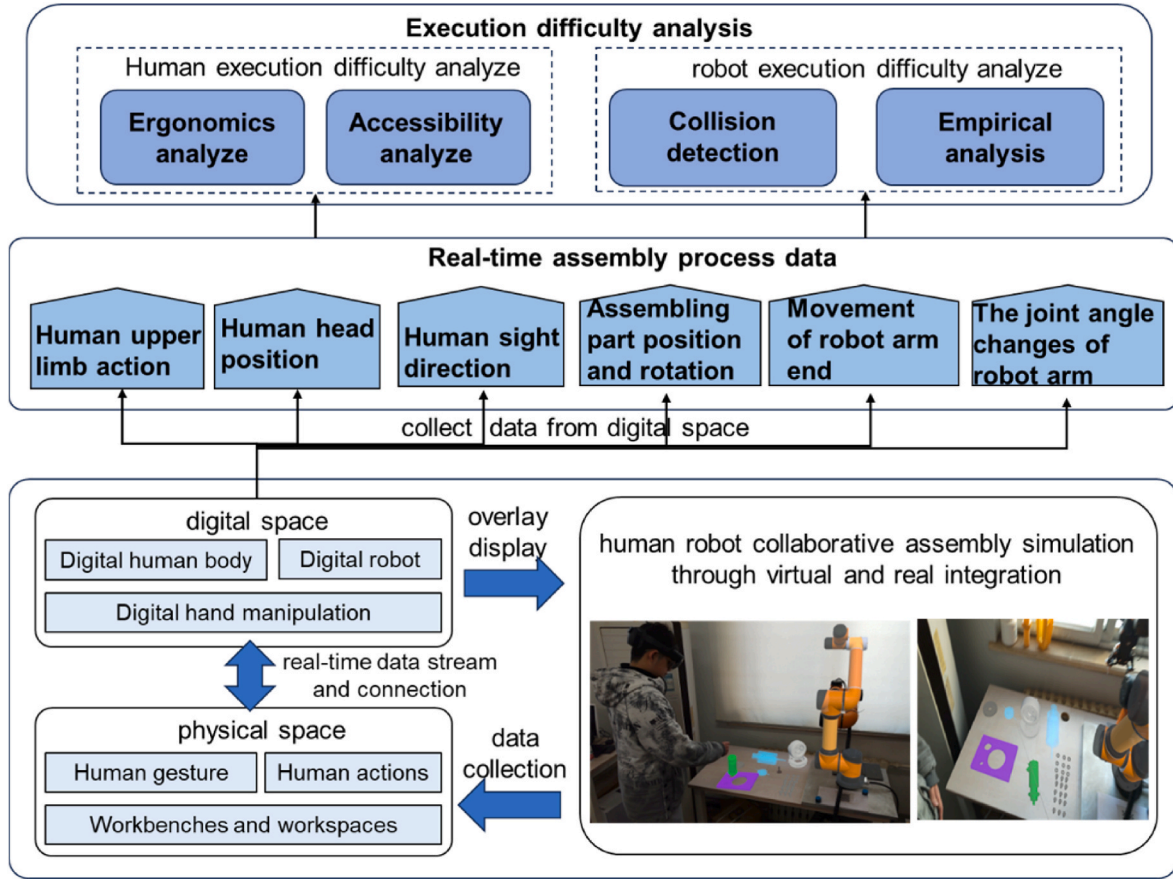


Fig. 3. The AR driven HRC assembly simulation process.

three sub-ergonomic indexes including Rapid Upper Limb Analysis (RULA), Ovako Working Posture Analysis (OWAS), and National Institutes of Occupational Safety and Health (NIOSH). These sub-ergonomic indexes are calculated by using the captured action records from human digital twin, and the method were reported in our previous research (Mao et al., 2023).

For robots, execution difficulty can mainly be divided into three aspects: motion accessibility, control complexity, and motion safety. The motion accessibility of robot is directly analyzed through robot control software, and then the value of motion accessibility index, MAI (can accessing,0; cannot accessing,1) is acquired. The control complexity index (OCI) is utilized to measure the complexity of robot controlling, and the safety index (SI) is used to evaluate the safety performance of robots performing subtasks. The final design of the execution difficulty for human and robot is given in Eq. (2) and Eq. (3):

$$\begin{cases} D_A^H i = \gamma_1 \bullet OAI + \gamma_2 \bullet EI & (\text{if } 0 < ORI, EI < 1) \\ D_A^H i = 1 & (\text{if } ORI = 1 \text{ or } EI = 1) \end{cases} \quad (2)$$

$$\begin{cases} D_A^R i = \varphi_1 \bullet MAI + \varphi_2 \bullet OCI + \varphi_3 \bullet SI & (\text{if } 0 < MAI, OCI, SI < 1) \\ D_A^R i = 1 & (\text{if } MAI = 1 \text{ or } OCI = 1 \text{ or } SI = 1) \end{cases} \quad (3)$$

Where $D_A^R i$ represents the difficulty of robot executing the installing operation of i -th task, and $D_A^H i$ represents the difficulty of human executing the installing operation of the i -th task. The γ_i and φ_i are coefficients, and $\sum_i^n \gamma_i = \sum_i^m \varphi_i = 1$. These coefficient values are also given by expert evaluation. Likewise, the difficulty of performing the installation operation $D_I^H i$, $D_I^R i$, and the difficulty of performing the tightening operation $D_T^H i$, $D_T^R i$, are evaluated and calculated in the same manner.

The control process of robot for an assembly task could be divided into three aspects: grasp, move and apply a force. Therefore, the OCI can be acquired by evaluating the difficulty indexes from the three parts and summing them. By drawing on the researches (Lamon et al., 2019; Tram and Raweevan, 2020), the evaluation methods for control complexity are established, as shown in Table 1.

For the safety of robots performing tasks, the collision avoiding between robots and humans is the main factor to be considered (see Table 2). The solution method for SI is as shown in Eq. (4).

$$SI = \frac{R}{N} \quad (4)$$

Where, N represents the length of the robotic arm's movement path, and

Table 1
the evaluation methods for control complexity.

Classification	Criteria	Score
Grasp	Simple grasp	0.2
	Need to grab with a specific posture	0.4
	The grab object is larger in shape and weight	0.6
	Grabbing requires replacement of special fixtures	0.8
	Cannot grasp the object	1.0
Move	Simple moving starting point to a destination point	0.2
	Obstacle avoidance is required during movement	0.4
	Multiple adjustments are required to move to a destination point	0.6
	Multiple adjustments are required to move to different destination points	0.8
	Cannot move to the destination point safely	1.0
Apply a force	Applying a simple force	0.4
	Applying forces in different directions	0.6
	Applying forces with resistance	0.8
	The applying force cannot meet the demand	1.0

Table 2
Multi-level randomization parameter table.

level	Variable	Range
Subtasks	N (Subtask number)	0–30
Single subtask i	C (Whether collaborative or not)	0 or 1
Execution data	D_A^H	0–1.0
	T_A^H	5–100 (s)
	D_A^R	0–1.0
	T_A^R	5–100 (s)
	D_I^H	0–1.0
	T_I^H	5–60(s)
	D_I^R	0–1.0
	T_I^R	5–60(s)
	D_T^H	0–1.0
	T_T^H	5–60(s)
	D_T^R	0–1.0
	T_T^R	5–60(s)

R represents the length of the robotic arm's movement within the human activity range.

4. Transfer reinforcement learning for task allocation method

4.1. The multi-agent transfer reinforcement learning process

The transfer reinforcement learning process is implemented based on the multi-agent proximal policy optimization (MA-PPO) and the knowledge distillation method. During the online training process, multiple agents (human agent and robot agent) interact with the environment in each episode to collect experience data including environment state, action, next state, and reward. Based on the experience data, the agents are updated by transfer multi-agent reinforcement learning

framework, as shown in Fig. 4. The entire multi-agent RL training process is conducted in a manner of centralized training and decentralized execution. Two agent processes the observation data that is observed from own view to take actions (A_H and A_R) by using the corresponding actor network. The central critic consists of two critic networks: human critic and robot critic, and these critic networks utilize the global state information to evaluate the value of state Q_R and Q_H . And for the next state (state'), the agents also use the same actor networks to take actions based on the next observations and the critic utilize the same critic networks to evaluate the next states giving the value of Q'_R and Q'_H . The TD-errors between two states are calculated to update the parameters of critic networks. At the same time, two teacher (teacher robot and teacher human) networks also give action suggestions (A_R^T and A_H^T) according to the observations in current state, and then the distillation loss evaluates the difference of actions and action suggestions. Besides, the PPO loss is calculated by integrating the Q-values given from critic networks with the Kullback-Leibler Divergence (KLD) of the sampling actor network and training actor network. These two kinds of loss are integrated to update the parameters of actor networks.

The framework is based on a completely decentralized learning strategy, and each agent is an independent PPO instance. Compared to the standard MAPPO, the state and reward design of the two agents is different, which is consistent with the actual situation of HRC system. Global information inputted to the central critic network improves the training efficiency, and local observation information is given to the action network to ensure that the learned strategy of agents can be executed autonomously without central guidance. In addition, using the teacher action network's suggestions to transfer accumulated experience is helpful to the actor networks, which tend to reduce the needed training samples and improve the training efficiency to achieve few-shot learning.

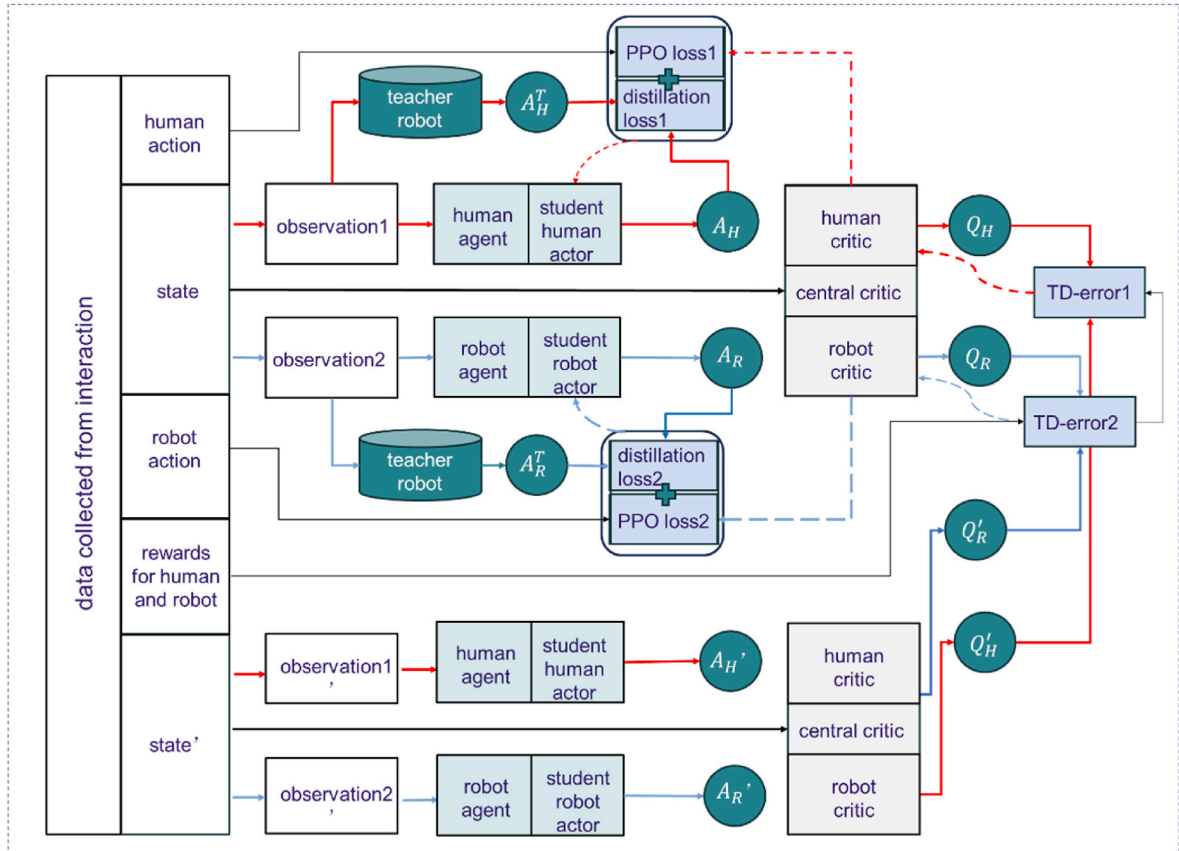


Fig. 4. The multi-agent transfer reinforcement learning framework for HRC task allocation.

4.2. The design of transfer multi-agent reinforcement

4.2.1. The state and observation space

The state information of the environment mainly includes the following categories: robot state S_R , human state S_H , and execution data E_i^j of the subtask $i+1$ to be assigned. The S_H and S_R are mainly used to describe the task execution status of humans and robots, involving the occupied state of the agent, working process, the current task number being executed or completed, and the current fatigue of human. For the state of robot, the S_R can be expressed as Eq. (5).

$$S_R = \begin{cases} (1, k = 1, i + 1, 0) & \text{finish the subtask, wait next} \\ & \text{subtask allocation} \\ (0, k, i, t) & \text{performing the subtask} \\ (0, k = 0, i, 0) & \text{perform the subtask collaboratively,} \\ & \text{waiting for collaborators} \end{cases} \quad (5)$$

Where the first value of array represents the busy state, and k represents whether the agent is performing a cooperative task. The sequence number of the execution subtask is represented by i , and t denotes the time required to execute the subtask. Compared with the robot state S_R , the human state S_H adds a fatigue index ε , which can be expressed in Eq. (6) according to literature (Glock et al., 2019).

$$\varepsilon = \sum_{i=1}^n 1 - e^{-\theta_i \cdot t_i} \quad (6)$$

In Eq. (5), the ε is the accumulated fatigue of human for n subtasks. The t_i is the execution time of the subtask i , and θ_i is a fatigue growth parameter when a worker executing the subtask i . This fatigue growth parameter involves the action's biomechanical analysis such as maximum strength in the required posture. Due to the difficulty of biomechanical parameter values obtaining, the difficulty index of human D_i^h is leveraged to approximate this fatigue growth parameter in our method. The D_i^h integrates Rapid Upper Limb Analysis, Ovako Working Posture Analysis, and National Institutes of Occupational Safety and Health, and these methods also analyze task execution difficulty from a biomechanical perspective. Therefore, the fatigue value of each subtask can be calculated by Eq. (7).

$$\varepsilon_i = 1 - e^{-D_i^h \cdot t_i} \quad (7)$$

Where the ε_i is the accumulated fatigue of human for subtask i . Furthermore, the fatigue state of human is obtained by accumulating the fatigue values of completed tasks.

In addition to the state data, the execution data is also needed for the task allocation decision. As mentioned below, the execution data consists of executing time and executing difficulty for working alone and working collaboratively. The execution data E_{i+1}^j is expressed in Eq. (1) of section 2.1.

The critic networks can acquire the global information $\{S_R, S_H, E_i\}$, while the agents utilize the local data to make decision: for the robot agent, the own state S_R and task execution data E_i can be observed; for the human agent, the own state S_H and E_i can be acquired.

4.2.2. The action space

For the task allocation problem of HRC, actions are the allocation results of subtasks. In the multi-agent RL process, the human agent and robot agent should take actions independently at the decision point. When an agent is in state of unoccupied, it is the decision point for making next subtask allocation decisions in MA-TRL. At each decision point, the human agent and the robot agent decide what actions to take for next subtask.

As mentioned in section 2.1, a subtask can be performed by human or robot independently or both in collaboration. In the case of collaborative execution of subtasks, the subtask can be divided into two parts: installation operation and fastening operation, each part needs to be performed independently by a human or robot. In order to adapt to the

neural network model, the action of an agent is represented as one-hot encodings, as shown in Eq. (8). Therefore, the output of the actor network is a vector consisting of four values, and the action selection is determined by sampling from this vector. The action A_j^h of the human agent is the same as that of the robot.

$$A_j^r = \begin{cases} (1, 0, 0, 0) & \text{Don't execute subtask } j \\ (0, 1, 0, 0) & \text{Execute subtask } i \text{ independently} \\ (0, 0, 1, 0) & \text{Cooperate to execute task subtask } j, \\ & \text{and perform the installation} \\ (0, 0, 0, 1) & \text{Cooperate to execute task subtask } j, \\ & \text{and perform the fastening} \end{cases} \quad (8)$$

Where j represents the subtask j that currently needs to be allocated, which means that allocation decisions are made for each subtask in sequence. When an agent is in state of "finish the subtask" (as shown in section 4.2.1), it is the decision point for making subtask allocation decisions. At each decision point, the agent of human actor and the agent of robot actor decide what actions to take for next task. Conflicting task allocation plans from these two agents will lead to the end of this episode of task allocation decision exploration.

4.2.3. The reward and loss functions design

During the RL algorithm training process, the agent of multi-agent RL tends to learn the action selection strategy with the maximum total reward. Therefore, the design of the reward function depends on the optimization goals of the RL algorithm. The optimizing goals of proposed task allocation method involves minimizing total tasks completion time, minimizing tasks execution difficulty and minimizing human fatigue. The factors representing different optimization objectives are integrated by weighted summation, and the weight coefficient are mainly decided to balance the order of magnitude difference between different factors. Therefore, the final reward functions are designed as Eq. (9) and Eq. (10).

$$r_H = -10 \cdot \varepsilon + \frac{10D + T}{T \cdot D} \quad (9)$$

$$r_R = \frac{10D + T}{T \cdot D} \quad (10)$$

Where ε is the fatigue of human that need to be minimized, and thus the minus sign is added. Because the rewards are calculated for single step, ε , D and T represents the execution fatigue, difficulty and the execution time for the current single step.

The goal of transfer reinforcement learning is to reused accumulated experience and meet the optimizing goals of new task scenario. Learning from accumulated experience is a soft goal because the agent is allowed not to fully adhere to the experience of the teacher network. The optimizing goal is a hard goal to guide the agent to learn the optimal allocation strategy for new task scenario. In order to unite the two goals, the loss functions of actor network in this MA-TRL include two parts: soft loss and hard loss, which can be expressed as Eq. (11).

$$\begin{cases} Loss^R = \mu_1 L_{Soft}^R + \mu_2 L_{Hard}^R \\ Loss^H = \pi_1 L_{Soft}^H + \pi_2 L_{Hard}^H \end{cases} \quad (11)$$

Where L_{Soft}^R and L_{Hard}^R represent the soft loss and hard loss for robot agent, while L_{Soft}^H and L_{Hard}^H represent the soft loss and hard loss for human agent. The soft loss is the knowledge distillation loss used to evaluate the difference between the student network action policy and the teacher network action policy. The hard loss is the PPO loss used to evaluate the current action policy by the state value from the current critic network. Besides, the value of weight coefficient μ_1 and π_1 are the hyper-parameters, and these parameters need to be set according to the actual learning process.

The knowledge distillation of transfer learning aims to transfer

accumulated experience to the actor network in the learning process. In order to realize the transfer of experience, it is necessary to calculate the difference between the output of the teacher action network and the output of the training action network in the same state. Since the output of the actor network is a probability distribution of actions, cross-entropy is used to measure the difference between these two distributions. Therefore, the soft loss, named distillation loss, is designed as Eq. (12).

$$\begin{cases} L_{Soft}^R = - \sum_i^N p_i^T \log(q_i^T) \\ p_i^T = \frac{A_i^{tea}/T}{\sum_{j=1}^n A_j^{tea}/T} \\ q_i^T = \frac{A_i^{stu}/T}{\sum_{j=1}^n A_j^{stu}/T} \end{cases} \quad (12)$$

Where L_{Soft}^R represent the distillation loss of robot agent, and N is the dimensions of the actions space. A_i^{tea} denotes the probability of teacher actor network taking action i at the state, and A_i^{stu} denotes the probability of training actor network taking action i at the same condition. The T is the hyperparameter of distillation temperature, and it should be set according to the transfer learning process.

For the critic network of human agent and robot agent, the TD-errors are used to update the network parameters, and the loss functions can be expressed as Eq. (13):

$$\begin{cases} Loss_c^H = \frac{1}{n} \sum_{i=1}^n [Q_H^i - (r_H^i + \gamma Q_H^{i+1})]^2 \\ Loss_c^R = \frac{1}{n} \sum_{i=1}^n [Q_R^i - (r_R^i + \gamma Q_R^{i+1})]^2 \end{cases} \quad (13)$$

Where the $Loss_c^H$ and $Loss_c^R$ represent the loss of human agent's critic network and loss of robot agent's critic network. The Q_H^i and Q_R^i are the value of state i by critic networks, while Q_H^{i+1} and Q_R^{i+1} are the value of next state $i+1$ by critic networks. The r_H^i and r_R^i are the rewards of human and robot in state i . The n represents the batch size of this train batch. The γ is a hyperparameter in algorithm.

4.3. Domain-randomization based pre-train for teacher networks

Before transfer learning, pre-training is required to obtain the teacher network. In order to learn useful experience and knowledge for real scenarios, many researches pay attention to establish methods to tackle the gap between the simulated task domain and the real task domain. Among these methods, the domain randomization is a widely used and effective technique, and this technique consists in training a DRL model over a variety of simulated environments with randomized parameters. For the purpose of enhance the effect of pre-training, the domain randomization is leveraged to train the teacher networks of robot agent and human agent. In the HRC task allocation problem, the differences in different scenarios mainly exist in changes in execution data caused by product and worker changing. Therefore, the randomized parameters chosen for domain randomized in training the teacher networks process including three levels: the subtasks, the single subtask and execution data. The utilized randomization parameters in this research are first presented as follows.

Except for the randomized parameters, the pre-training also utilized the MA-PPO framework and environment that are same as transfer training so that the experience of the teacher networks can adapt to the transferred agents. In the training process, execution data is randomly sampled to construct the training environment in each episode. The number of subtasks is first randomly determined, and then each

parameter of every subtask is determined by randomly sampling from the range. The training should be terminated until the total rewards converges or the maximum number of iterations. The total reward curve of domain randomization training is shown in Fig. 5.

4.4. The update of the network

For the transfer learning process, the networks updating follow the online learning due to the extended framework of MA-PPO. Different from the pre-training, the environment parameters in this transfer reinforcement learning process are collected by the digital twin system. In the training process, the agents will take actions and be rewarded in the environment to generate training samples, and then these samples are used to update the networks. In the updating process, the network of each PPO instance is updated separately according to its respective loss. Unlike the traditional PPO method, in addition to the PPO losses, the distillation losses are also calculated by evaluating the gap between the outputs of the teacher actor network and the current actor network for same states, as presented in step 10 and 11. The parameters of the actor networks are update by weighted sum of PPO losses and distillation losses.

Algorithm:

Input: assembly subtasks execution dataset H, discount factor γ , η , learning rate lr , distillation temperature T, the current networks of actor, the current networks of critic, the maximum iterations T, the network update times E.

1. Initialize all network parameters.
2. for episode = 1 to T do:
3. Initialize S_R, S_H, E_i to be the first state of the current state sequence, get eigenvector $\phi(S), \phi(Ob1)$ and $\phi(Ob2)$
4. While (if subtasks are all finished) {
5. Current actor networks get the actions $a_R = \pi(\phi(Ob1))$, $a_H = \pi(\phi(Ob2))$
6. Execute action and observe reward R, observe new state S' and eigenvector $\phi(S') \phi(Ob1')$ and $\phi(Ob2')$
7. Record the $\phi(S), \phi(Ob1), \phi(Ob2), (a_H, a_R), (r_H, r_R), \phi(S'), \phi(Ob1'), \phi(Ob2')$, and is end in to sample queue }
8. For $i = 1$ to E:
9. Calculate the PPO losses through calculating KLD of the sampling actions and actions of current actor networks for same states
10. Calculate the distillation losses by evaluating the gap between the outputs of the teacher actor network and the current actor network for same states
11. Calculate the value losses of the current critic networks by timing difference
12. Update the parameters of the current actor networks by weighted sum of PPO losses and distillation losses
13. Update the parameters of the current critic networks by value losses

Output: time series of tasks of Agent-Human and Agent-Robot

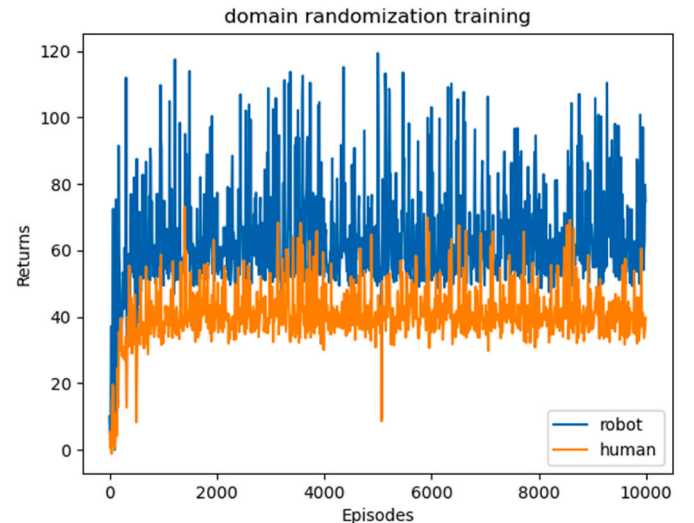


Fig. 5. The total reward curve of domain randomization training process.

5. Case study

5.1. Assembly case

The virtual assembly process of a key component from the real automatic guide vehicle (AGV) assembly line is used to test and verify the method proposed. As shown in Fig. 6, the AGV steering wheel assembly includes 18 key parts and many standard fastening parts such as bolts and screw. The numbers of key parts represent the assembly sequence, so there are 18 assembly subtasks of this steering wheel assembly.

Based on this assembly case, our proposed method is test and validated for task allocation of HRC. As mentioned before, the execution data of each subtask needs to be collected firstly through the AR simulation platform. The collected execution data are then used as environment parameters to train the proposed multi-agent transfer reinforcement learning (MA-TRL) model. In order to compare and verify the proposed algorithm, the rule-based method, the heuristic-based algorithm, the algorithm based on improved multi-agent deep deterministic policy gradient (MA-DDPG) (Zhang et al., 2022) and the algorithm based on multi-agent deep-Q-network (MA-DQN) (Yu et al., 2021) are also utilized to solve the task allocation problem for comparison. In addition, to further illustrate the advantages of the proposed algorithm, three hypothetical working condition variations of this assembly case were utilized to test and compare the adapting performance of the three algorithms. Finally, an actual HRC assembly experiment in a real scenario was performed to verify the results of the algorithm decision-making.

5.2. Comparative experiment and results analysis

In order to obtain the execution data of assembly subtasks of AGV steering wheel assembly, an assembly operation simulation is performed through an AR and digital twin-based assembly simulation platform. As shown in Fig. 7, the digital twin-based assembly simulation platform includes a virtual six-axis robot of physical six-axis robot equipped with camera, the operator wearied with AR headset, virtual parts and containers, and a real workbench. In order to obtain the execution data of assembly subtasks of AGV steering wheel assembly, assembly operation simulation is performed through this assembly simulation platform. For each subtask, three kinds of situations, which includes human executing alone, robot executing alone, and human-robot collaborative executing, are all considered and simulated. In the situation of human-robot collaborative assembly, two kinds of possible process are taken into account: the part is installed by the operator first and then tightened by robots; the part is installed by robot first and then tightened by the operator. In the assembly simulation process, the execution time and difficulty of each situation are evaluated through the indexes proposed

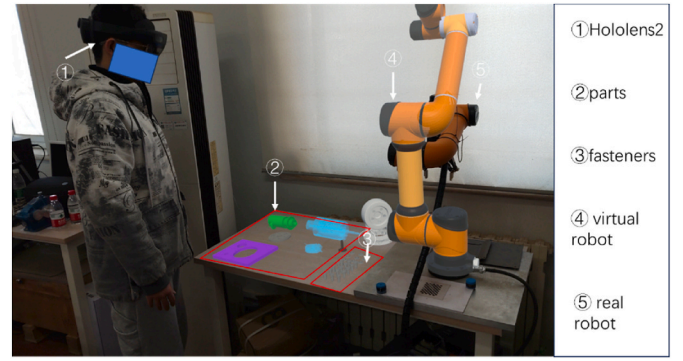


Fig. 7. The AR and digital-twin based assembly simulation platform.

in section 3.2.

Ten students, ranging in age from 22 to 30 and from mechanical engineering school, were recruited to participate in the experiment. They performed HRC assembly simulations after being trained on the use the system. The obtained execution data of each subtask from 10 users are averaged, as shown in Table 3. Each value is defined in section 4.2.1, and the blank values in the table means that an agent cannot process the subtask independently or the subtask cannot be performed collaboratively. In addition, these values of execution difficulty only retain one significant figure.

In the experiment, the execution data are used as RL training environment parameters, and the proposed MA-TRL, improved MA-DDPG and MA-DQN are all trained in the same environment to solve this HRC task problem. In addition, rule-based methods and heuristic-based algorithm are also used to solve the same problem for comparison. For rule-based methods, the minimum execution time rule is applied in this comparative experiment, in which the subtasks are allocated in a scheme with minimum executing time. Since the task allocation problem is a multi-objective problem, the non-dominated sorting genetic II (NSGA-II) algorithm as the represent of heuristic-based algorithm is used in this experiment. In the NSGA-II algorithm, minimizing the total work time, minimizing the average execution difficulty, and minimizing total human fatigue are used as optimization objectives. In the parameter setting of these RL algorithms, the all networks are all designed as a fully connected network with a hidden layer of 128 nodes; and the hyper-parameters are presented in Table 4. The model is trained on a computer equipped with an AMD Ryzen 7945HX CPU, an NVIDIA GeForce RTX 4060 GPU, and 16 GB of memory. The deep learning framework used is pytorch 2.1.1. For NSGA-II, the iteration is set to 1000 generations; the population size is set to 200; the mutation distribution parameter and the crossover distribution parameter are both set to 1. In addition, the result of NSGA-II algorithm is randomly selected from the pareto solution set.

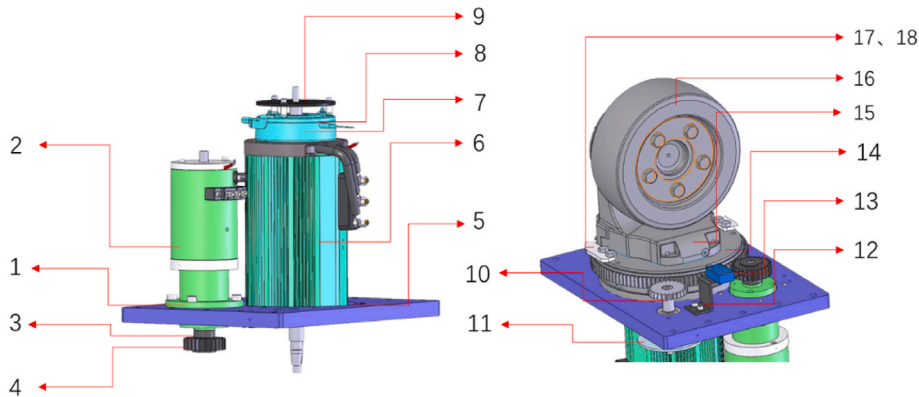


Fig. 6. The structure of the AGV steering wheel.

Table 3

The execution data of the subtasks of AGV steering wheel assembly.

Task number	D_A^H	D_A^R	T_A^H	T_A^R	D_I^H	D_I^R	T_I^H	T_I^R	D_T^H	D_T^R	T_T^H	T_T^R
1	0.5	0.8	30	56	0.4	0.5	15	25	0.5	0.8	20	38
2	0.5	0.6	24	18	0.5	0.6	16	10	0.2	0.2	10	8
3	0.5	1	20									
4	0.6	0.9	66	54								
5	0.4	0.5	24	15								
6	0.7	0.9	45	49	0.7	0.9	35	30	0.4	0.5	12	8
7	0.5	0.5	25	32	0.4	0.5	12	10	0.5	0.5	12	25
8	0.6	1	42		0.5	0.6	30	24	0.6	1	18	
9	0.6	0.8	63	48	0.4	0.3	15	10	0.6	0.6	55	40
10	0.4	0.5	31	20	0.3	0.4	18	10	0.4	0.5	12	12
11	0.4	0.2	46	42	0.2	0.4	12	14	0.4	0.4	40	28
12	0.6	0.8	36	48	0.4	0.8	25	18	0.6	0.6	30	26
13	0.8	1	48		0.6	0.8	20	10	0.8	1	28	
14	0.4	0.4	18	12								
15	0.6	0.9	75	42	0.3	0.8	26	22	0.6	0.6	53	24
16	0.5	1	59		0.4	0.3	12	15	0.6	1	28	
17	0.6	1	36									
18	0.6	1	36									

Table 4

The parameters of the RL-based methods.

algorithms	learning rate	discount Factor	batch size	soft update parameters	T	μ_1	μ_2	π_1	π_2
MA-TRL	0.003	0.98	32	–	7	0.4	0.6	0.4	0.6
MA-DQN	0.003	0.98	32	–	–	–	–	–	–
MA-DDPG	0.003	0.98	32	0.005	–	–	–	–	–

The solution results of these algorithms are shown in Table 5. In the results of the rule-based method, the total execution time is the smallest, but human fatigue and task execution difficulty are largest, which shows that using the rule cannot obtain multi-objective optimization results. The result of NSGA is similar to the result of MA-DQN, in which the task allocation is not balanced, resulting in greater working difficulty and fatigue. The MA-DDPG and MA-TRL can obtain the best results between different optimization goals. In addition, compared with the RL based methods, the solution time of the NSGA-II method is significantly longer. Compared with MA-DDPG, the robot execution difficulty in the results of MA-TRL is slightly larger, but the total human fatigue and execution time are smaller. The results show that the proposed MA-TRL method can obtain the best task allocation solution compared with other methods. Since the experience of the pre-trained model is referred to through knowledge distillation, the proposed MA-TRL is not prone to fall into the local optimal situation. The RL-based methods can greatly improve the solution speed after training compared to heuristic algorithms.

The training effectiveness of the proposed MA-TRL is compared with other RL-based algorithms through the RL reward curve of the training process, and the reward curves are as shown in Fig. 8. As can be seen from Fig. 8, the convergence speed of the proposed MA-TRL is much greater than that of improved MA-DDPG and MA-DQN. In addition, the convergence stability of MA-TRL is also the best among the three. The

results show that the proposed transfer reinforcement learning can improve the training process and significantly increase the training speed.

In actual manufacturing, the working conditions often change, resulting in changes in execution data. Three hypothetical situations are utilized to change the working conditions, and then to change the execution data. Three situations include: a check action is added after some subtask tightened is completed, the execution time of the subtask will increase; Changing the placement location of parts, so the robot or human cannot access the parts and the execution difficulty is 1; the new subtask of parts moving is added which could be executed by the robot or human alone. Based on these situations, three working conditions were designed in addition to the original working conditions, as the shown in Table 6.

Based on training under initial working conditions, the three RL algorithms were directly used to make decisions on HRC task allocation under different working conditions. Additionally, three RL algorithms were retrained for each work conditions, and then used to make decisions. In addition, the rule-based method and NSGA-II algorithm were also utilized in each condition for comparison. The parameters of the algorithm are consistent with the above comparative experiments, and the results comparisons are shown in Tables 7–9. In the tables, the blank data values in the table indicate that a complete decision-making solution cannot be obtained due to errors in decision-making, for example,

Table 5

The results of different algorithms.

algorithm	training time (s)	solution time (s)	The comparison of solution result				
			Makespan (s)	Total work time of human and robot (s)	Total Fatigue of human	Average task difficulty of human	Average task difficulty of robot
Rule	–	0.001	694	392,302	13.90	0.54	0.63
NSGA-II	–	313.352	718	690,28	13.36	0.53	0.40
MA-DQN	226.63	0.0453	721	610,111	14.12	0.52	0.50
MA-DDPG	453.89	0.034	702	518,198	13.48	0.52	0.51
MA-TRL	26.55	0.039	690	441,249	12.41	0.49	0.55

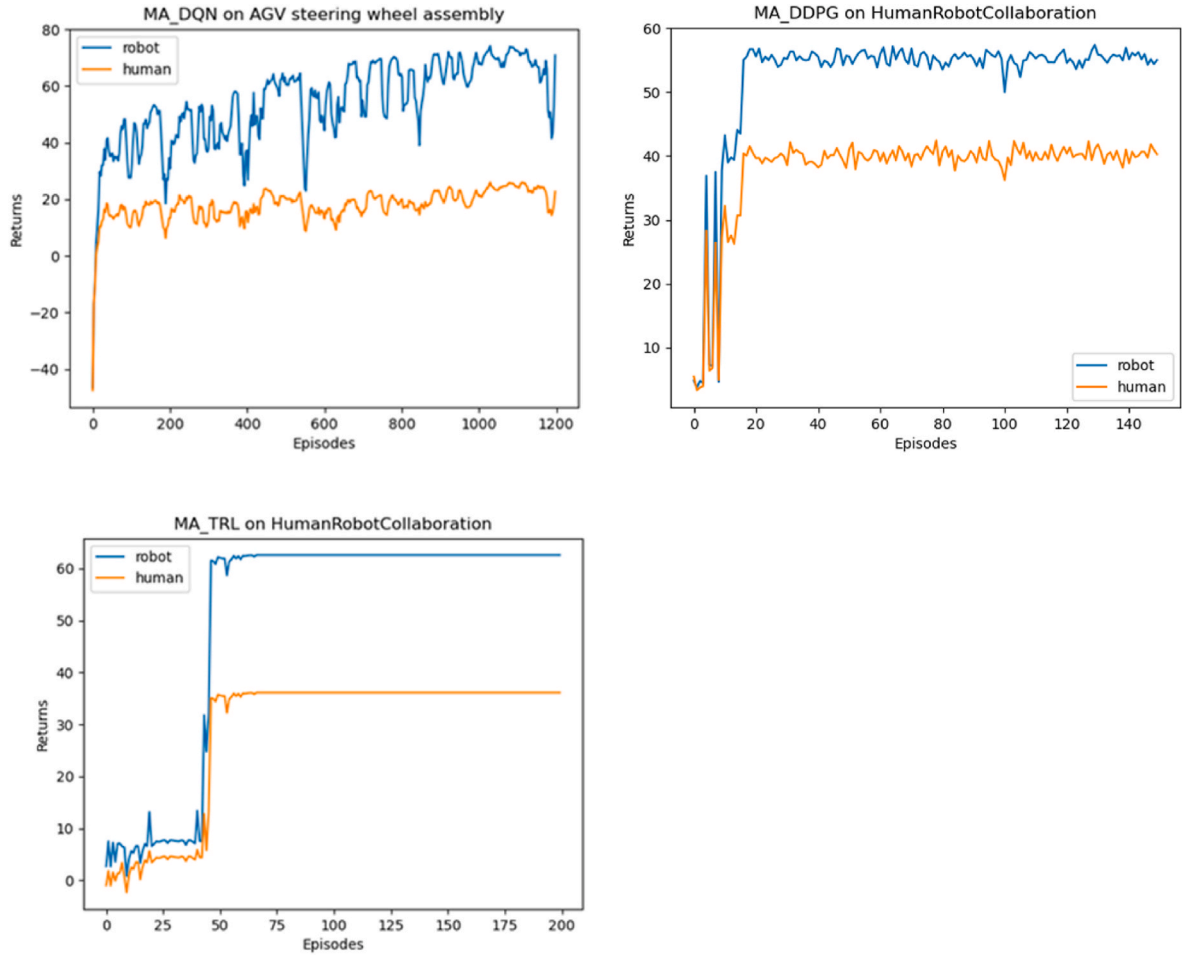


Fig. 8. (a) The total reward curves during MA-DQN training. (b) The total reward curves during improved MA-DDPG training. (c) The total reward curves during MA-TL training.

Table 6

Execution data modified contents of different work conditions.

Name	Changed value numbers	Changed content
Assembly condition 0 (A0)	0	Original working condition, the execution data as shown in Table 3;
Assembly condition 1 (A1)	16	Change the execution time of subtask2, subtask 7, subtask 12 and 17 by adding check action;
Assembly condition 1 (A2)	26	On the basis of A1, set the execution difficulty of subtask 13, subtask 14, and subtask 15 by setting the parts place to make the robot cannot access
Assembly condition 3 (A3)	38	On the basis of A2, add 3 subtasks of parts moving before the subtask 3, the subtask 10, the subtask 16;

assigning task to agent that cannot perform or conflicting task assignments.

In training conditions, the “Initial trained” means that the model trained under the original assembly condition is used directly to solve the problem, and “Retrained” means that the model is retrained under the current assembly condition and then used to solve the problem.

The results show that the retrained RL models can obtain better task allocation solutions than the RL models trained with historical experience and rule-based method in new conditions. The retraining time of the proposed algorithm is significantly less than that of other MARL-

based algorithms. Although the NSGA-II based method can also solve the optimal allocation solutions, the solution speed of the NSGA-II method is significantly slower than that of the RL-based method. Moreover, the retraining time for the proposed method is notably less than the solution time required by the metaheuristic algorithm. When working conditions change significantly, the model trained based on historical data may not even be able to find a feasible solution. The results illustrate that when working conditions change, the algorithm model needs to be adjusted and retrained to learn more accurate strategies. The results illustrate that the proposed MA-TL algorithm greatly improves the speed of retraining, which is very helpful when facing large-scale decision-making problems with changeable conditions.

5.3. Decision-making results validation

In order to verify the results of algorithmic decision-making, the three from above-mentioned participants also conducted real HRC assembly experiments in real scenarios. According to the HRC task allocation plan solved by MA-TL, MA-DDPG and MA-DQN, the users and the robot performed the HRC task of the assembly case in original working condition, as shown in Fig. 9. The execution time data of the user and the robot were collected, and the Gantt chart were made based on these real data and plan data from simulation, as shown in Fig. 10. The results show that although there is a small gap between the actual task execution time and the solved execution time based on the assembly simulation data, the solved solutions are all feasible in actual assembly. Besides, as seen from Fig. 10 and Table 10, in the real assembly experiment, the task allocation scheme solved by MA-TL has a more

Table 7

The results of different algorithms under assembly condition A1.

algorithm	training conditions	Training times to converge	Training time (s)	Solution time (s)	Makespan (s)	Total fatigue of human	Average difficulty of human	Average difficulty of robot
MA-DQN	Initial trained	–	–	0.0230	768	13.08	0.52	0.73
	Retrained	1000	243.41	0.0170	772	12.41	0.48	0.52
MA-DDPG	Initial trained	–	–	0.0378	781	13.51	0.52	0.58
	Retrained	1400	379.52	0.0420	740	13.36	0.52	0.53
MA-TRL	Initial trained	–	–	0.0330	735	13.41	0.57	0.63
	Retrained	90	14.26	0.0360	726	12.93	0.51	0.57
Rule	–	–	–	0.0012	634	13.91	0.54	0.63
NSGA-II	–	–	–	358.49	752	14.32	0.53	0.50

Table 8

The results of different models under assembly condition A2.

algorithm	training conditions	Training times to converge	Training time (s)	Solution time (s)	Makespan (s)	Total fatigue of human	Average difficulty of human	Average difficulty of robot
MA-DQN	Initial trained	–	–	0.0250	–	–	–	–
	Retrained	1100	297.36	0.0217	799	14.13	0.53	0.48
MA-DDPG	Initial trained	–	–	0.0330	811	12.03	0.51	0.68
	Retrained	1400	433.97	0.0371	797	12.57	0.49	0.51
MA-TRL	Initial trained	–	–	0.0315	788	13.07	0.52	0.54
	Retrained	80	12.44	0.0381	789	12.70	0.50	0.51
rule	–	–	–	0.0010	658	13.26	0.51	0.60
NAGA-II	–	–	–	459.07	764	15.10	0.54	0.40

balanced results between multiple optimal goals of working difficulty, working time and human fatigue, compared to the schemes solved by the other two algorithms.

5.4. Discussion

In the comparative experiment of the case study, the multi-agent algorithm MA-DDPG and MA-TRL were able to solve for better results compared to rule-based methods and those based on the metaheuristic algorithm NSGA-II. The game between multiple agents during training can make the task allocation between robots and workers more balanced. Compared with other multi-agent reinforcement learning algorithms, the proposed MA-TRL method has faster training speed and better training stability. Furthermore, the three advanced MARL algorithm is applied to the task allocation problem under different working

conditions. The results show that that the trained MARL models are difficult to find the optimal solution or even a feasible solution when the working conditions change greatly. These MARL models need to be retrained to adapt to changing task allocation problems. During the retraining process, the results show that the proposed MA-TRL algorithm can greatly improve the convergence speed compared with other MARL algorithms. This advantage could save algorithm training costs, which is crucial when facing large-scale decision-making problems. In addition, although the proposed MA-TRL is the online-learning based algorithm, it can obtain the same optimal results as the offline algorithm of improved MA-DDPG. This indicates that the MA-TRL does not get trapped in local optimal results like other online learning algorithms because fully trained teacher experience is added in training.

If the RL model is retrained for new working condition, the execution data of new working condition must be obtained to construct a new

Table 9

The results of different models under assembly condition A3.

algorithm	training conditions	Training times to converge	Training time (s)	Solution time (s)	Makespan (s)	Total fatigue of human	Average difficulty of human	Average difficulty of robot
MA-DQN	Initial trained	–	–	–	–	–	–	–
	Retrained	1100	–	0.0302	864	16.77	0.53	0.45
MA-DDPG	Initial trained	–	–	–	–	–	–	–
	Retrained	1500	542.76	0.0356	855	16.78	0.53	0.42
MA-TRL	Initial trained	–	–	0.0275	864	16.77	0.53	0.51
	Retrained	100	17.24	0.0332	836	16.02	0.52	0.53
Rule	–	–	–	0.00104	783	17.53	0.50	0.60
NSGA-II	–	–	–	540.715	846	17.03	0.53	0.50

**Fig. 9.** HRC assembly experiment.

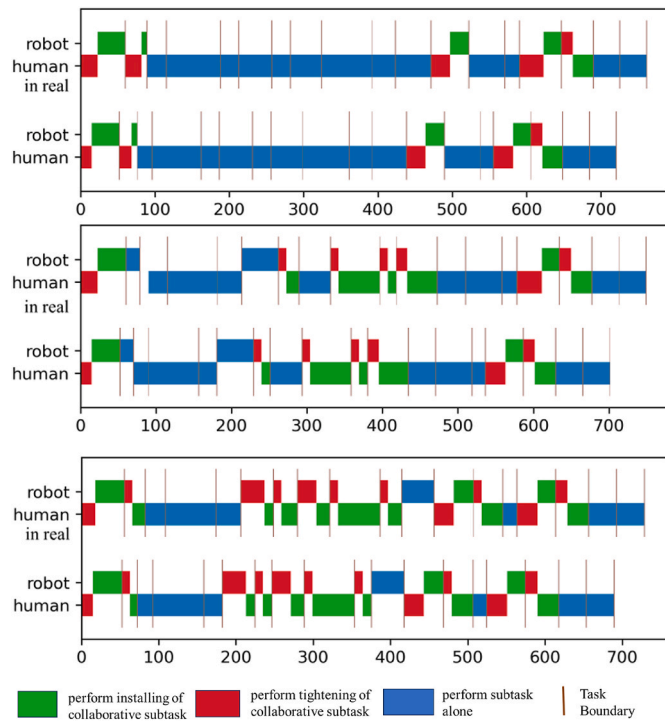


Fig. 10. (a) the Gantt chart of recorded and simulation-based plan subtask execution time of task allocation scheme from MA-DQN; (b) the Gantt chart of recorded subtask execution time of task allocation scheme from MA-DDPG; (c) the Gantt chart of recorded subtask execution time of task allocation scheme from MA-TRL.

Table 10

The working results of the HRC task allocation schemes solved by three algorithms.

Scheme	Total working time (s)		Average working difficult		Total fatigue
	Robot	Human	Robot	Human	
MA-DQN	111	651	0.50	0.52	14.56
MA-DDPG	188	562	0.53	0.52	13.76
MA-TRL	249	470	0.481	0.517	12.89

target training environment. Based on our proposed AR and digital-twin based simulation platform, execution data can be obtained in a low cost and efficiently, because the data could be automatically collected and processed and physical product prototype are not required. For the assembly case, the simulation experiment was performed to obtain execution data. Based on these data, three algorithms were trained to solve their respective task allocation schemes. A real HRC assembly experiment was performed to verify and compare these schemes. The results of real experiments show that there is a small gap between the actual task execution time and the planned execution time by applying simulation data. The gap is mainly in human execution time, causing by the uncertainty of human task execution and the manual assembly simulation based on gesture interaction. Since the virtual robot simulation based on the teaching program of real robot, the gap between the real robot and virtual robot simulation in executing task is very small. The results illustrate that effective initial task allocation planning solutions can be obtained by applying the proposed method. Furthermore, the HRC task allocation scheme can be further optimized based on real execution time data by using the proposed MA-TRL.

6. Conclusion

Currently, multi-agent reinforcement learning methods can well

solve the complex task allocation problem in human-robot collaborative assembly. However, due to the variability of real assembly working condition, there are obstacles in the application of learning methods. The newly changed working conditions and scenes lack enough data to support decision model training, and already trained models need to be retrained to adapt to new working conditions. In order to solve this gap to apply multi-agent reinforcement learning methods in practical problems, this paper proposes a transfer reinforcement learning and digital-twin based task allocation method. Based on the digital twin and AR assembly simulation of human-robot collaboration system, an assembly execution difficulty evaluation method is proposed to collect assembly execution data to support MA-RL model training. Integrating knowledge distillation and multi-agent PPO algorithms, a multi-agent transfer reinforcement learning framework is proposed to use execution data to efficiently train accurate decision-making models through knowledge reusing. Based on the case of AGV steering wheel assembly, the performance of the proposed method is test and verified. Based on the execution data collected by the AR simulation experiment, the training process and result of the algorithm was compared with current advanced MA-RL methods including MA-DQN and MA-DDPG in different working conditions. The results show that using the proposed method can find the optimal task allocation scheme, and the training efficiency of this method is much improved, which is crucial in the large-scale decision-making problems. Furthermore, a real HRC assembly experiment is performed to demonstrate the effectiveness of the proposed method.

With the application of human-robot collaboration in manufacturing systems, the proposed method can be utilized to effectively solve the task allocation problem between operators and robots under dynamic working conditions. However, although the real-time human actions and robot actions can be obtained by digital-twin connection in this method, the acquisition of real-time state data of the real assembling parts is limited. Therefore, the HRC execution data is obtained using AR assembly simulation to replace real physical assembly, which still has a certain gap with the real assembly data. The assembly task allocation scheme solved based on simulation execution data can be used as a feasible initial scheme. In the future, the state capturing method of digital-twin needs to be improved to obtain real-time data from the actual HRC system. The real-time data will be utilized to continuously improve the task allocation scheme through rapid solving and training of MA-TRL. Besides, the proposed method mainly considers the task allocation within a single human-robot collaborative assembly workstation, which is not suitable for the scenario of multi-operator and multi-robot collaboration. In the future, the algorithm will be improved and applied to complex multiple agent collaboration scenarios.

CRedit authorship contribution statement

Jingfei Wang: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Yan Yan:** Supervision, Methodology. **Yaoguang Hu:** Writing – review & editing, Supervision, Conceptualization. **Xiaonan Yang:** Writing – review & editing, Supervision. **Lixiang Zhang:** Writing – review & editing, Validation.

Declaration of generative AI in scientific writing

During the preparation and writing of this work all the authors didn't use the generative AI.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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